Nael Fessha

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EDUCATION

Georgia Institute of Technology — Atlanta, GA
Bachelor of Science in Electrical Engineering | GPA: 3.30 — *Expected May 2028*
Relevant Coursework: Programming for HW/SW, Digital Design, Differential Equations

Georgia State University — Atlanta, GA
Bachelor of Science in Computer Science | GPA: 3.74 — *Aug 2024 – May 2025*
Awards/Honors: President's List | Relevant Coursework: Python

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, Python, HTML, C, RISC-V Assembly
Software: Excel, PowerPoint, Word, Power BI, GitHub, Arduino IDE, KiCad 9.0
Hardware/Tools: FPGA design, Logic circuits, Multimeters, Oscilloscopes, Voltmeters, Transformers

WORK EXPERIENCE

Georgia Institute of Technology — Atlanta, GA
Student Assistant — *Jan 2025 – Present*
- Coordinated seating arrangements for 100+ students per exam session, ensuring an efficient testing environment
- Ensured testing accommodations for 20+ students with documented disabilities, coordinating seating, timing, and materials

Chick-fil-A — Duluth, GA
Shift Leader — *Apr 2023 – Aug 2023*
- Led training for 5 new team members on front-end procedures and customer service to integrate them into the team
- Provided exceptional service contributing to a 93% customer satisfaction rating

PROJECTS

Embedded Sidescroller Game with Custom Hash Table *(C, ESP32-C6, FreeRTOS)*
- Engineered a 2D side-scroller on the ESP32-C6 microcontroller with a 128×128 uLCD interface, integrating real-time input handling, GPIO-driven audio, and dynamic sprite rendering
- Implemented a custom hash table with separate chaining for spatial lookup of game objects, enabling O(1) tile queries across a cyclic 60×9 world map with procedural panel generation
- Optimized drawing pipeline with anti-flicker caching and frame-budgeted rendering for 10+ simultaneous entities on resource-constrained hardware

Pathfinding Optimization Engine *(C, RISC-V Assembly)*
- Engineered an iterative, non-recursive solution using 5-bit binary masking as a hardware-level multiplexer to evaluate 128 path permutations without branch overhead
- Optimized spatial and temporal performance at the ISA level, minimizing dynamic execution length, static code size, and operand storage against strict baseline metrics

Temperature Controlled Fan *(C/C++, Arduino IDE)*
- Designed an Arduino-based temperature monitoring system using a DHT11 sensor and PWM-controlled fan to regulate cooling in real time
- Integrated LCD display output, sensor data acquisition, and motor driver control in a closed-loop embedded system

LEADERSHIP AND PROFESSIONAL DEVELOPMENT

Management Leadership for Tomorrow — Atlanta, GA
Career Preparation Fellow — *Sep 2025 – Present*
- Accepted into a selective 20-month professional development program for high-achieving talent
- Complete business case studies and assignments to grow leadership and technical skills

Caleb Foundation - Save Life with Pennies — Atlanta, GA
Lead Volunteer — *Aug 2019 – Present*
- Volunteer at concession stands on weekends to support a free daycare program in Ethiopia serving 110 children
- Raised $22,640 by recruiting friends and family in my community to attend fundraisers and volunteer

ADDITIONAL SKILLS AND INTERESTS

Interested in Embedded Systems, Circuit Debugging, entrepreneurship, and international travel.